

Evaluation of Three Formulations of *Beauveria bassiana* for Control of Lesser Mealworm and Hide Beetle in Georgia Poultry Houses

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ABSTRACT Initial screening of 12 *Beauveria bassiana* (Balsamo) Vuillemin isolates against larvae of the lesser mealworm (*Alphitobius diaperinus* [Panzer]) resulted in the selection of two isolates, GH4 and 707, for further testing under field conditions. Three formulations of each strain were prepared: an EC, a ground corn granular formulation, and waste product of fungal propagation containing spent media, mycelia, and unharvested conidia ("residue" formulation). Two field trials were conducted in commercial caged-layer houses in Georgia with 5–6 mo of manure accumulation and established populations of *A. diaperinus* and hide beetles (*Dermestes maculatus* DeGeer). In the first trial field, *B. bassiana* was applied a single time to the manure surface at either 10^9 (EC and granular cornmeal bait formulations) or 10^8 (residue formulation) fungal spores per square meter. In the second trial, two successive weekly treatments were applied, using a total of 6X the rate of application used in the first trial. Significant treatment effects were short-lived and only detected 2 wk after treatment in both trials. The granular formulations of both strains and the residue formulation of the GH4 strain provided the greatest degree of suppression (60–90%) of beetle larvae. A laboratory bioassay confirmed that the granular bait was the most effective formulation. More frequent applications made earlier in the manure accumulation cycle may be necessary to achieve satisfactory control of these beetles.

KEY WORDS *Alphitobius diaperinus*, *Dermestes maculatus*, *Beauveria bassiana*, poultry

THE LESSER MEALWORM, *Alphitobius diaperinus* (Panzer), and hide beetle, *Dermestes maculatus* DeGeer, are among the most important pests of the poultry industry throughout the world (Safrit and Axtell 1984, Turner 1986, Geden et al. 1999). *A. diaperinus* is a reservoir of numerous avian and human pathogens, and their larvae are highly destructive pests in building insulation and other soft construction materials (McAllister et al. 1994, 1996; Vaughan et al. 1984; Despins et al. 1994; LeTorché and Letenneur 1983). Damage to building materials occurs when larvae reach critical population levels in the litter or manure, resulting in dispersal of mature larvae seeking pupation sites (Ichinose et al. 1980, Geden and Axtell 1987). Hide beetles are also commonly found in poultry houses, where they feed on cracked chicken eggs, dead birds and rodents, and manure itself (Cloud and Collison 1986, Stafford et al. 1988). Hide beetle larvae damage buildings in a manner similar to that of *A. diaperinus* and, in addition, also tunnel into structural timbers to the point of compromising the soundness of building supports. Annual economic losses to these insects, known collectively as litter beetles, have been

estimated at \$16 million and \$10 million in Virginia and Georgia, respectively (Turner 1986, Riley et al. 1997).

Because insecticidal control of litter beetles is generally unsatisfactory and short-lived, there is a need for alternative management approaches. Mechanical barriers can be used to prevent beetles from emigrating from the manure to reach susceptible building components (Geden and Carlson 2001), but this approach does not limit populations living in the manure. The beetles have several natural enemies, including mites and protozoans (Husband and Baker 1992, Steinkraus and Cross 1993, Steinkraus et al. 1992, Apuya et al. 1994, Bala et al. 1990). Lesser mealworms are also susceptible to steinernematid and heterorhabditid nematodes, which can provide short-term control in the field (Geden et al. 1985, 1987, Geden and Axtell 1988).

In previous studies we found that larvae, pupae, and adults of *A. diaperinus* were susceptible to the entomopathogenic fungus *B. bassiana*, and that there were substantial differences in virulence among fungal strains (Steinkraus et al. 1991, Geden et al. 1998). Although direct treatment of infested poultry litter and manure is not the most effective delivery method, this is a simple application method that is likely to be used by commercial producers. The objective of the current study was to determine the effectiveness of

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manure treatments with different strains and formulations of *B. bassiana* for control of established litter beetle populations in commercial poultry houses.

Materials and Methods

Strain Selection Bioassay. Twelve isolates of *B. bassiana* maintained by Mycotech Co. (Butte, MT) were screened for virulence against larvae of *A. diaperinus* in the laboratory: 706, 707, 708, 709, 711, 712, GHA, 1201a, 1201b, 1202a, 1202b, 1202c. Spores were harvested from agar plates of live culture material, then spore counts per mg of collected spores were determined by hemacytometer counts. Spore counts ranged from 1.2×10^7 (isolate 1202c) to 1.2×10^8 (isolate 706) spores/mg.

Spore viability was determined for each isolate by diluting 1 mg of spores in 1 ml of distilled water containing 0.01% Tween 80, making a 1:10 dilution, then plating 0.1 ml on a petri dish containing SDA agar (two dishes/isolate). After 24 h, four coverslips were placed on each plate over a drop of lactophenol-acid fuschin stain, and the number of germinated and ungerminated spores in one field of view under each coverslip was determined at 400 $\times$. Spore viability ranged from 62.0 to 99.1%.

Spores were weighed and mixed with corn starch to prepare two concentrations of viable spores for each isolate; 1.5×10^7 and 1.5×10^9 spores/g of starch. Forced-contact bioassays were then conducted separately in our laboratories in Arkansas and Florida. Spore mixtures were tested against mature *A. diaperinus* larvae by exposing groups of 10 larvae to 0.1-g aliquots of the test mixtures per group. After a 30-s period of contact, the larvae were removed from the inocula with soft forceps and transferred to clean petri dishes lined with filter paper premoistened with distilled water. Cornstarch with no spores was used for controls. Larvae were monitored for mortality on days 3, 5, and seven after exposure. The bioassays were replicated either four (Arkansas) or five times (Florida), using beetle larvae collected from poultry houses in Arkansas and Georgia, respectively. Data from the two laboratories were pooled and two isolates were selected for production, formulation, and field evaluation.

Field Trials in Georgia. Three formulations of the two selected isolates (GHA and 707) were prepared by Mycotech for field-testing. The first was an EC formulation containing 2×10^{11} spores/ml. The second was a coarsely ground granular corn bait consisting of 8.45×10^9 spores/g. The third formulation, "residue," was a waste product of fungal propagation comprised of spores and mycelia remaining in spent fungal growth medium after automated spore harvesting at Mycotech's production facility. This product, which was processed to the consistency of coarse cornmeal, contained 6.5×10^8 spores/g.

Field tests were conducted at a commercial poultry farm near Hoboken, GA, consisting of four layer houses containing $\approx 120,000$ birds each. In the first trial, 30 ≈ 10 -square-meter plots of manure in one of

the houses were delineated by placing flags along the sections of manure ≈ 2 -m wide by 5 m long. Manure had accumulated for ≈ 6 mo at the beginning of the study. Two tube traps (Safrit and Axtell 1984) were placed in each plot and adult and larval beetles of *A. diaperinus* and *D. maculatus* were collected from the traps and counted for three weeks before assigning treatment groups to the plots. The two plots with the lowest larval *A. diaperinus* counts were dropped, leaving 28 plots to allocate among the six fungal treatments and controls (four plots/treatment). Plots were allocated by first ranking them by counts of *A. diaperinus* larvae and grouping them into four sets of seven plots (i.e., plots no. 1-7 in abundance of beetle larvae, 8-14, etc.), in descending order of larval abundance. Treatments were assigned randomly within each group of seven plots until all 28 plots had been assigned. By doing this, plots were assigned to treatments so that there were no significant pretreatment differences in *A. diaperinus* larval densities among the treatment groups.

B. bassiana was applied to the manure plots on 31 August, 2000, at 10^9 spores per square meter for the EC and granular formulations. The residue formulation was applied at a lower dose (10^8 spores per square meter) because of the lower spore concentration inherent in this product. The EC was applied by diluting 50 ml of concentrated spore suspensions in 1000 ml of water for each plot and spraying the manure with a calibrated CO₂ sprayer. The two dry formulations were applied by hand with conventional restaurant grated cheese shakers. Beetle populations were monitored weekly with tube traps as before for 3 wk after treatment.

A second trial was conducted in a different poultry house at the Hoboken farm in September–November, 2000. Manure in the house had accumulated for ≈ 5 mo at the start of testing. Methods were the same as described above, except that two applications of *B. bassiana* were made, 1 wk apart, at three times the spore dose used in the first trial. Also, pretreatment counts were made for 4 wk before allocation to treatment groups. Treatments were applied on 26 October and 2 November. Beetle populations were monitored weekly for 3 wk after the second treatment.

Data were analyzed by one-way analysis of variance (ANOVA) separately for each week after performing log ($x + 1$) transformations of trap counts, and treatment means were separated using the REGWQ algorithm of the GLM Procedure of the Statistical Analysis System (SAS Institute 1998).

Laboratory Bioassays. Laboratory assays were conducted in Arkansas using the same rates of application used in the first field trial (six fungal treatments plus controls). In these tests, groups of 100 field-collected larvae were first placed in screen-topped 3-liter plastic tubs containing 800 cm³ of poultry litter from a broiler house. Litter was frozen before testing to kill any insects present at the time of collection. Treatments were applied to the litter surface 1-h after addition of the larvae and the tubs held at 25°C for three weeks. Chicken feed was added to the tubs as needed and

Table 1. Relative virulence of 12 strains of *B. bassiana* for field-collected mature lesser *A. diaperinus* larvae after a brief forced-contact exposure to spores mixed with corn starch

Strain	Mean (SE) percent mortality of <i>A. diaperinus</i> after exposure to <i>B. bassiana</i>			
	Day 5 after exposure to dose		Day 7 after exposure to dose	
	1.5 × 10 ⁶	1.5 × 10 ⁸	1.5 × 10 ⁶	1.5 × 10 ⁸
1201a	10.0 (10.0)	33.5 (10.5)	67.0 (33.0)	95.0 (5.0)
1201b	14.5 (2.5)	60.5 (12.5)	28.5 (18.5)	77.0 (23.0)
1202a	8.5 (5.5)	36.5 (1.5)	54.0 (16.0)	83.0 (17.0)
1202b	5.5 (2.5)	41.0 (3.0)	58.5 (24.5)	86.0 (14.0)
1202c	6.5 (3.5)	16.0 (6.0)	84.0 (16.0)	97.0 (3.0)
706	49.0 (49.0)	100.0 (0.0)	96.5 (3.5)	100.0 (0.0)
707	60.0 (40.0)	100.0 (0.0)	100.0 (0.0)	100.0 (0.0)
708	3.5 (0.5)	95.0 (5.0)	38.5 (18.5)	99.0 (1.0)
709	6.0 (6.0)	76.5 (0.5)	60.5 (26.5)	98.5 (1.5)
711	5.0 (5.0)	10.0 (10.0)	20.5 (13.5)	60.5 (2.5)
712	4.0 (4.0)	9.0 (5.0)	61.5 (31.5)	80.0 (20.0)
GHA	13.5 (9.5)	23.5 (13.5)	63.0 (37.0)	87.0 (13.0)
control	5.0 (5.0)	5.0 (5.0)	10.0 (10.0)	10.0 (10.0)

Doses are spores per 0.1 g of corn starch used in the assays.
N = 4 (Arkansas) and 5 (Florida) replications, 10 larvae per strain and dose.
Results are means of tests replicated in Florida using Georgia beetle populations and in Arkansas using local beetle populations.

moisture levels maintained by periodically measuring moisture loss from the tubs and adding water to compensate. Larvae were removed by hand from one-third of the tubs at 1, 2, and 3 wk after treatment and numbers of live and dead larvae counted (N = 4 tubs/treatment/time interval for a total of 84 tubs). Data were analyzed by ANOVA and treatment means separated weekly as before.

Results and Discussion

All of the isolates tested were virulent for *A. diaperinus* larvae in forced-contact bioassays (Table 1).

The low concentration of 1.5 × 10⁷ spores/g of corn starch was selected because this was the approximate LC₅₀ observed using this bioassay method in a previous *B. bassiana* evaluation against this pest (Geden et al. 1998). The 707 isolate was the most virulent in these tests (100% mortality at the low dose on day 7 and 100% mortality at the high dose on day 5) and was, therefore, selected for the field tests. The GHA strain ranked near the middle of the isolates tested in virulence for beetle larvae; however, it was included in the field trials because this isolate possesses desirable characteristics with respect to production yield, harvesting efficiency, and storage stability. Moreover, a product containing this isolate (Mycotrol) is already registered for use against whiteflies, aphids, thrips, mealybugs, leafhoppers, weevils, and leaf-feeding insects in a variety of agricultural systems.

In the first field trial, the only significant treatment effects on *A. diaperinus* larvae were observed at 2 wk after treatment (Table 2), when the granular formulations of both isolates and the residue formulation of GHA resulted in >50% reduction in the number of larvae collected in tube traps (means of 15.5–18.3 larvae/trap) compared with the controls (40.3 larvae/trap) (F = 3.32; df = 6,21; P = 0.019). This suppression was short-lived, however, and no significant treatment effects were observed on week 3. Variation in adult *A. diaperinus* counts was high and prevented detection of significant treatment effects at P = 0.05 (Table 2). However, as was observed with the larvae of this species, there were numerically fewer adult beetles collected from plots treated with the two granular formulations and the residue formulation of GHA (means, 28.0–43.3 adults/trap) compared with the control plots (129.8 adults/trap).

Hide beetle populations, although relatively small, were affected by the treatments in much the same way as *A. diaperinus* (Table 3). Treatment effects on

Table 2. Mean (SE) *A. diaperinus* per tube trap before and after manure treatment with *B. bassiana* in trial no. 1

Weeks posttreatment	<i>B. bassiana</i> strain 707 formulation			<i>B. bassiana</i> strain GHA formulation			Control
	EC	Granular	Residue	EC	Granular	Residue	
<i>A. diaperinus</i> larvae							
−3	5.5 (2.9)a	3.0 (1.1)a	5.5 (1.0)a	5.8 (2.8)a	7.0 (2.3)a	7.0 (3.2)a	2.8 (2.1)a
−2	4.0 (1.1)a	9.0 (5.0)a	17.0 (7.0)a	6.0 (2.5)a	10.3 (4.9)a	18.5 (5.6)a	2.3 (0.5)a
−1	30.3 (10.1)a	35.5 (19.5)a	16.8 (5.3)a	25.5 (7.0)a	22.0 (11.6)a	13.3 (4.4)a	32.5 (11.4)a
Pre-trt mean	13.2a	15.8a	13.1a	12.4a	13.1a	12.9a	12.5a
0	12.5 (4.3)a	27.0 (8.8)a	55.3 (28.1)a	49.3 (18.3)a	7.3 (2.2)a	41.0 (8.8)a	38.3 (7.2)a
1	26.8 (12.9)a	31.3 (12.6)a	18.8 (3.8)a	62.0 (14.6)a	8.0 (2.9)a	69.5 (27.3)a	26.0 (6.0)a
2	24.5 (5.4)ab	15.5 (5.5)b	33.5 (10.6)ab	59.3 (8.0)a	18.0 (5.1)b	18.3 (4.3)b	40.3 (15.6)a
3	31.0 (6.0)a	51.3 (24.8)a	59.5 (39.7)a	29.5 (6.1)a	27.0 (6.7)a	38.3 (15.7)a	29.5 (8.9)a
<i>A. diaperinus</i> adults							
−3	36.0 (8.6)a	37.3 (16.0)a	42.3 (8.5)a	25.8 (7.3)a	64.0 (30.8)a	33.0 (8.1)a	37.5 (11.7)a
−2	49.3 (33.4)a	38.8 (14.2)a	78.5 (20.3)a	35.8 (13.2)a	32.0 (11.8)a	84.8 (48.1)a	31.5 (12.8)a
−1	150.5 (66.5)a	105.3 (28.9)a	85.3 (22.3)a	91.8 (39.3)a	72.5 (24.5)a	90.0 (40.8)a	131.8 (19.8)a
Pre-trt mean	78.6a	60.5a	68.7a	51.1a	56.2a	69.2a	66.9a
0	144.8 (73.7)a	93.0 (58.1)a	92.3 (31.5)a	113.3 (40.5)a	32.3 (18.0)a	70.8 (21.9)a	33.5 (4.5)a
1	104.5 (60.3)a	58.3 (24.9)a	50.8 (7.0)a	68.5 (16.9)a	20.3 (9.7)a	80.5 (17.5)a	31.0 (12.2)a
2	86.0 (31.5)a	43.3 (9.4)a	89.8 (49.2)a	83.8 (24.3)a	28.0 (4.6)a	37.0 (8.9)a	129.8 (75.8)a
3	150.3 (92.6)a	73.0 (28.5)a	234.0 (183.4)a	66.5 (22.9)a	42.3 (2.3)a	117.0 (71.3)a	73.3 (22.1)a

Treatments applied on 31 Aug. (week 0) to 10 m² manure plots at 10⁹ (EC and granular) or 10⁸ (residue) *B. bassiana* spores per m². N = 4 plots/treatment, two tube traps/plot. Means within rows followed by the same letter are not significantly different at P = 0.05 (REGWQ).

Table 3. Mean (SE) *D. maculatus* per tube trap before and after manure treatment with *B. bassiana* in trial no. 1

Weeks post-treatment (date)	<i>B. bassiana</i> strain BB-707 formulation			<i>B. bassiana</i> strain GHA formulation			Control
	EC	Granular	Residue	EC	Granular	Residue	
<i>D. maculatus</i> larvae							
−3	11.5 (3.7)a	12.0 (4.6)a	13.8 (4.2)a	24.5 (10.2)a	11.0 (4.0)a	10.5 (2.2)a	10.5 (5.5)a
−2	4.5 (2.6)a	8.0 (3.1)a	8.0 (1.3)a	15.8 (6.6)a	3.8 (2.8)a	4.8 (1.3)a	8.8 (3.7)a
−1	24.8 (10.3)a	33.0 (13.0)a	20.0 (5.8)a	40.8 (13.2)a	13.8 (5.4)a	12.5 (3.1)a	34.8 (17.0)a
Pre-trt mean	13.6a	17.7a	13.9a	27.0a	9.5a	9.3a	18.0a
0	18.3 (7.4)a	6.3 (1.6)a	10.5 (3.4)a	23.0 (6.8)a	8.3 (5.0)a	5.3 (2.9)a	19.5 (16.8)a
1	8.5 (2.5)a	7.5 (2.9)a	9.8 (6.3)a	22.0 (7.2)a	5.8 (2.3)a	5.5 (0.9)a	12.8 (11.4)a
2	15.3 (3.7)a	2.0 (0.0)bc	5.5 (2.3)ab	10.8 (5.7)ab	1.3 (0.9)c	1.3 (0.3)c	11.0 (6.4)a
3	10.8 (6.3)a	1.0 (0.7)a	2.3 (0.9)a	7.5 (3.1)a	3.5 (1.3)a	5.3 (3.0)a	7.0 (3.4)a
<i>D. maculatus</i> adults							
−3	0.5 (0.5)b	1.5 (0.6)ab	2.8 (0.6)ab	8.8 (3.7)ab	3.3 (2.3)ab	2.3 (1.0)ab	13.5 (5.7)a
−2	1.0 (0.7)a	1.8 (1.2)	3.3 (0.9)a	7.3 (3.7)a	1.0 (0.6)a	2.0 (0.7)a	4.3 (2.0)a
−1	5.5 (1.8)a	12.0 (9.4)a	7.0 (6.0)a	18.8 (13.5)a	2.8 (1.5)a	5.5 (2.2)a	3.8 (2.2)a
Pre-trt mean	2.3a	5.1a	4.4a	11.6a	2.4a	3.3a	7.2a
0	6.5 (3.0)a	1.8 (0.9)a	1.5 (0.6)a	8.0 (5.2)a	1.3 (1.3)a	2.3 (0.9)a	4.5 (3.6)a
1	4.3 (1.4)a	2.5 (1.2)a	3.5 (1.4)a	8.0 (4.6)a	1.3 (0.8)a	3.3 (1.7)a	5.3 (4.6)a
2	4.3 (1.3)a	1.5 (1.0)ab	0.5 (0.3)ab	2.3 (0.5)ab	0.0 (0.0)b	0.0 (0.0)b	1.0 (1.0)ab
3	5.5 (2.9)a	0.0 (0.0)b	0.0 (0.0)b	2.8 (0.9)ab	0.0 (0.0)b	2.8 (2.1)b	3.0 (1.2)ab

Treatments applied on 31 Aug. (week 0) to 10 m² manure plots at 10⁹ (EC and granular) or 10⁸ (residue) *B. bassiana* spores per m². N = 4 plots/treatment, two tube traps/plot. Means within rows followed by the same letter are not significantly different at P = 0.05 (REGWQ).

larvae were only significant on week 2 after treatment, with the granular formulation (both isolates) and GHA residue plots having 12–18% as many larvae as the control plots (*F* = 5.15, *df* = 6,21; *P* = 0.002). Treatment effects on hide beetle adults were erratic, presumably because of the small numbers collected.

In the second field trial, populations of *A. diaperinus* were higher overall than in the first trial (Table 4). The only significant treatment effects on larvae were ob-

served on week 1 after the second treatment (2 wk after the first fungal application). As in the first trial, the two granular formulations and the GHA residue plots had significantly lower beetle populations (mean, 41–57.3 larvae/trap) compared with the controls (144.0 larvae/trap) (*F* = 6.36, *df* = 6,21; *P* = 0.001), but the suppression was short-lived. No significant treatment effects were observed among *A. diaperinus* adults (Table 4) or hide beetle adults or larvae in the second trial (Table 5).

Table 4. Mean (SE) *A. diaperinus* per tube trap before and after manure treatment with *B. bassiana* in trial no. 2

Weeks post-treatment (date)	<i>B. bassiana</i> strain BB-707 formulation			<i>B. bassiana</i> strain GHA formulation			Control
	EC	Granular	Residue	EC	Granular	Residue	
<i>A. diaperinus</i> larvae							
−4	8.8 (1.4)a	10.3 (3.2)a	15.5 (3.2)a	10.5 (4.1)a	16.5 (9.0)a	7.8 (1.5)a	2.8 (0.8)a
−3	15.0 (3.3)a	5.0 (1.4)a	10.5 (1.0)a	4.5 (2.1)a	23.8 (12.8)a	5.8 (4.1)a	11.0 (4.0)a
−2	19.0 (8.2)a	7.0 (2.9)a	15.5 (6.3)a	3.5 (0.6)a	6.0 (3.7)a	3.8 (1.9)a	10.8 (4.2)a
−1	46.5 (18.2)a	49.8 (7.2)a	33.5 (3.7)a	42.0 (9.1)a	41.8 (7.9)a	60.0 (12.9)a	42.0 (10.2)a
Pre-trt. mean	22.3a	18.0a	18.8a	15.1a	22.0a	19.3a	16.6a
0	36.3 (7.0)a	62.3 (16.0)a	24.5 (4.4)a	51.8 (21.8)a	66.0 (19.4)a	20.5 (6.5)a	35.0 (5.7)a
0	16.8 (4.8)ab	35.0 (28.0)ab	7.0 (1.4)ab	15.3 (5.7)ab	2.8 (1.3)b	11.3 (0.6)ab	38.3 (18.5)a
1	74.5 (9.3)abc	57.3 (10.0)bc	86.5 (12.8)abc	116.0 (30.8)ab	41.0 (4.7)c	47.8 (8.0)c	144.0 (13.4)a
2	33.3 (4.8)a	40.3 (12.3)a	22.5 (6.6)a	44.8 (13.3)a	37.8 (13.0)a	15.8 (3.8)a	82.5 (39.3)a
3	23.0 (14.2)a	40.5 (16.7)a	13.3 (5.5)a	25.3 (13.3)a	40.0 (16.6)a	6.8 (2.6)a	68.8 (30.0)a
<i>A. diaperinus</i> adults							
−4	31.3 (5.6)a	23.8 (6.9)a	34.3 (8.6)a	38.3 (11.5)a	31.5 (7.0)a	32.0 (10.4)a	32.3 (6.0)a
−3	51.8 (24.5)a	37.0 (6.6)a	93.5 (51.1)a	36.5 (8.1)a	45.8 (16.9)a	23.8 (7.4)a	74.8 (26.1)a
−2	21.8 (6.2)a	12.5 (3.0)a	64.0 (40.9)a	21.5 (1.9)a	22.3 (7.7)a	15.8 (4.8)a	34.5 (15.3)a
−1	14.5 (5.0)a	61.8 (26.7)a	46.3 (21.4)a	66.8 (40.0)a	69.5 (33.6)a	60.8 (33.6)a	57.8 (39.0)a
Pre-trt. mean	29.9a	33.8a	59.5a	40.8a	42.3a	33.1a	49.9a
0	62.8 (8.4)a	86.5 (23.6)a	70.8 (44.8)a	142.0 (107.9)a	58.0 (17.2)a	92.0 (25.2)a	143.3 (44.1)a
0	82.5 (7.3)a	110.3 (33.0)a	110.8 (19.0)a	189.5 (119.7)a	168.0 (90.7)a	192.3 (18.5)a	147.8 (42.0)a
1	187.5 (53.8)a	229.8 (112.7)a	191.3 (29.0)a	202.8 (48.8)a	148.3 (30.5)a	225.8 (44.7)a	343.8 (89.4)a
2	174.8 (48.1)a	178.5 (75.7)a	125.0 (38.4)a	97.8 (16.5)a	252.0 (56.3)a	113.3 (58.2)a	237.5 (46.4)a
3	160.8 (77.2)a	96.5 (46.6)a	36.0 (6.0)a	54.5 (41.9)a	148.0 (139.4)a	57.0 (29.0)a	172.3 (80.4)a

Treatments applied on two consecutive weeks (26 Oct. and 2 Nov.) to 10 m² manure plots at 3 × 10⁹ (EC and granular) or 3 × 10⁸ (residue) *B. bassiana* spores/m². N = 4 plots/treatment, 2 tube traps/plot. Means within rows followed by the same letter are not significantly different at P = 0.05 (REGWQ).

Table 5. Mean (SE) *D. maculatus* per tube trap before and after manure treatment with *B. bassiana* in trial no. 2

Weeks post-treatment	<i>B. bassiana</i> strain BB-707 formulation			<i>B. bassiana</i> strain GHA formulation			Control
	EC	Granular	Residue	EC	Granular	Residue	
<i>D. maculatus</i> larvae							
−4	1.3 (1.3)	0.8 (0.5)	2.3 (1.7)	0.8 (0.5)	6.3 (5.9)	2.5 (1.0)	0.0 (0.0)
−3	0.3 (0.3)	0.3 (0.3)	1.0 (0.6)	0.5 (0.5)	4.0 (2.7)	0.5 (0.5)	0.3 (0.3)
−2	1.5 (0.9)	1.5 (0.6)	2.5 (1.0)	1.0 (0.6)	8.8 (6.8)	3.5 (2.2)	3.3 (1.6)
−1	0.3 (0.3)	2.3 (1.1)	1.8 (0.9)	1.0 (0.6)	2.8 (1.5)	0.8 (0.3)	1.0 (0.4)
Pre-trt. mean	0.9	1.2	1.9	0.8	5.5	1.8	1.2
0	3.5 (1.0)	2.5 (0.3)	7.8 (3.7)	4.3 (1.7)	3.8 (2.8)	4.0 (0.4)	7.8 (5.5)
0	5.3 (3.1)	9.3 (1.8)	7.8 (3.3)	5.8 (3.2)	4.0 (2.7)	10.3 (2.6)	6.8 (2.3)
1	7.0 (4.0)	7.5 (3.2)	8.3 (3.1)	5.3 (2.1)	6.8 (2.2)	6.5 (1.3)	8.5 (2.2)
2	7.3 (2.0)	4.8 (3.4)	10.0 (4.1)	4.5 (0.9)	12.0 (4.3)	7.0 (2.1)	8.8 (4.4)
3	4.0 (2.1)	6.0 (5.7)	2.3 (0.5)	5.3 (1.8)	1.8 (0.9)	3.3 (1.4)	4.0 (0.9)
<i>D. maculatus</i> adults							
−4	0.0 (0.0)	0.0 (0.0)	1.3 (0.6)	0.0 (0.0)	3.8 (3.8)	0.0 (0.0)	0.0 (0.0)
−3	0.0 (0.0)	0.3 (0.3)	0.3 (0.3)	0.3 (0.3)	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)
−2	0.0 (0.0)	0.3 (0.3)	1.8 (1.4)	0.5 (0.3)	4.0 (3.7)	0.5 (0.5)	0.5 (0.5)
−1	0.3 (0.3)	1.3 (0.6)	0.0 (0.0)	0.0 (0.0)	0.3 (0.3)	0.5 (0.3)	0.3 (0.3)
Pre-trt. mean	0.1	0.5	0.9	0.2	2.0	0.3	0.2
0	1.0 (1.0)	1.5 (1.2)	2.8 (1.8)	2.8 (2.1)	1.5 (0.6)	4.5 (2.2)	9.0 (8.3)
0	2.8 (2.4)	6.5 (2.9)	3.3 (2.0)	6.0 (3.5)	4.5 (3.1)	12.0 (8.1)	1.8 (1.1)
1	3.8 (2.1)	8.5 (6.0)	5.5 (3.0)	4.3 (2.5)	7.5 (5.2)	4.0 (2.3)	7.0 (2.9)
2	3.3 (2.0)	5.0 (3.0)	3.5 (1.7)	7.0 (1.7)	3.3 (1.1)	2.5 (2.2)	22.0 (19.0)
3	0.5 (0.5)	1.5 (1.5)	0.0 (0.0)	0.8 (0.5)	0.5 (0.5)	1.5 (1.2)	1.8 (0.9)

Treatments applied on two consecutive weeks (26 Oct. and 2 Nov.) to 10 m² manure plots at 3 × 10⁹ (EC and granular) or 3 × 10⁸ (residue) *B. bassiana* spores/m². N = 4 plots/treatment, 2 tube traps/plot). There were no significant treatment effects on any week at P = 0.05 (REGWQ).

These field trials represented rather strenuous test conditions for the fungus. The manure pack was already large (5–6 mo accumulation) and robust populations of *A. diaperinus* were already present at the beginning of the trials. Given these circumstances and the rapid burial of the applied spores under a constantly falling carpet of fresh manure deposits, our tests may be seen as representing a worst-case scenario for *B. bassiana* poultry house applications. The effects of manure conditions such as pH, moisture, and feed additives on fungal efficacy are unknown. Nonetheless, the results indicate that reductions in larval densities of both *A. diaperinus* and hide beetles can be achieved with fungal treatments and that granular and residue formulations are more effective in poultry manure than an EC formulation. Further support for

this is provided by results of the laboratory experiment, where the granular formulations provided greater control than the EC (Table 6).

In summary, the results of the current study indicate that single or two weekly treatments of *B. bassiana* were not effective in providing long-term control of established resident litter beetle populations in a large manure mass. However, the results are sufficiently encouraging to warrant further evaluations under test conditions more favorable to the fungus. For example, weekly treatments during the first few weeks after manure clean out would increase the probability of the fungus impacting the target population in time to prevent the beetles from reaching outbreak levels. Incorporation of food attractants or semiochemicals into granular baits or waste residue products could improve the efficacy of these materials. Carefully timed applications of infective baits could also be integrated with other bio- and behavior-based management approaches such as mechanical barriers, mites, protozoans, and fungal treatments directed at migrating larvae (Bala et al. 1990, Steinkraus et al. 1992, Steinkraus and Cross 1993, Apuya et al. 1994, Geden et al. 1998, Geden and Carlson 2001).

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Table 6. Mortality of *A. diaperinus* larvae in laboratory arenas treated with *B. bassiana* using 10⁹ (EC and granular) or 10⁸ (residue) *B. bassiana* spores/m²

Strain	Formulation	Mean (SE) percent mortality of <i>A. diaperinus</i> larvae after exposure to <i>B. bassiana</i> treatment		
		Week 1	Week 2	Week 3
Control		5.1 (1.8)a	15.8 (3.0)c	34.3 (5.0)c
707	EC	3.8 (0.5)a	28.1 (4.0)bc	45.2 (7.1)c
707	Granular	5.2 (1.6)a	57.2 (3.2)a	81.5 (6.6)a
707	Residue	4.6 (2.7)a	33.8 (4.2)b	48.7 (4.0)bc
GHA	EC	7.8 (2.9)a	26.8 (4.3)bc	41.5 (4.4)c
GHA	Granular	6.8 (3.1)a	28.5 (8.5)bc	65.9 (9.8)ab
GHA	Residue	8.1 (1.8)a	19.3 (5.0)c	48.1 (5.7)bc

N = 4 arenas containing 800 cm³ of used broiler litter and an initial population of 100 *A. diaperinus* larvae. Means within columns followed by the same letter are not significantly at P = 0.05 (REGWQ).

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